

Sagnik Roy

Hooghly, West Bengal – 712149

Phone: +91 7076404580

Email: sagnikroyofficial24@gmail.com

LinkedIn: <https://www.linkedin.com/in/sagnik-roy-b39218192/>

GitHub: <https://github.com/Sagnikroy12>

PROFESSIONAL SUMMARY

SDET with 3 years of experience building scalable UI/API automation frameworks and high performance automation systems using Playwright, Selenium, REST Assured, Python and TypeScript. Specialized in automation architecture, workflow optimization, browser automation and reducing large-scale manual operations through engineering focused tooling. Strong experience in CI/CD integration, API testing, microservices validation and automation-driven process acceleration.

CORE SKILLS

- **Automation:** Playwright (Typescript), Selenium WebDriver, REST Assured, Karate, Postman
- **Languages:** Java, Python, TypeScript
- **Frameworks:** TestNG, JUnit, Cucumber (BDD), POM, Data Driven Frameworks.
- **Testing:** API, UI, Microservices, Cross-Browser, Parallel Execution.
- **Tools:** Git, GitHub, Azure DevOps, GitHub Actions, Docker (Basics), JIRA
- **Data & ML:** Pandas, NumPy, OCR(Tesseract), Image processing, ML-assisted validation

EXPERIENCE

Senior Software Engineer (A5) — Capgemini, Hyderabad | Jan 2025 – Present

- Built ML-assisted validation systems achieving ~95% coverage and reducing defects by 30%.
- Engineered a reconciliation automation engine that reduced manual processing time from 1 week to under 1 minute.
- Led AI-driven automation initiatives improving release stability and reducing QA cycles by ~25%
- Designed scalable automation workflows integrating validation, reporting, and data pipelines.

Software Engineer (A4) — Capgemini, Hyderabad | Apr 2023 – Jan 2025

- Developed reusable API automation frameworks for data validation, reporting workflows, and API verification pipelines.
- Implemented Cucumber BDD for business-readable automation and improved test maintainability
- Optimized regression execution through parallel execution strategies, reducing execution time by 30%.
- Performed end-to-end testing across UI and API layers in Agile environments.

PROJECTS

IRCTC Ticket Automation System (Playwright + TypeScript)

- Engineered a high-speed Playwright based automation system optimized for Tatkal ticket booking workflows.
- Implemented human-like interactions simulation and CAPTCHA automation handling for anti bot resistant workflows.
- Integrated ML-based CAPTCHA solver for real-time decoding.
- Built a GUI-based runtime control system for dynamic passenger, station and payment configuration.

Recon Engine —Automation Tool (Python)

- Built a modular reconciliation system for large scale financial dataset validation and discrepancy detection.
- Implemented hybrid matching logic using exact and fuzzy matching for real world discrepancy detection.
- Implemented large dataset processing using pandas vectorization techniques.
- Generated automated validation reports with Pandas vectorization techniques.

Resume Builder Web Application

- Developed and deployed a resume generation platform with dynamic template rendering and structured formatting.
- Designed scalable data-to-document transformation workflows for ATS-friendly resume generation.
- Deployment: (<https://aipoweredresumebuilder.netlify.app>)

EDUCATION

Bachelor of Mechanical Engineering — Academy of Technology, MAKAUT | GPA: 8.54 | Jun 2022

CERTIFICATIONS

- SAlFe 6 Practitioner
- API Testing Using REST Assured
- Webservices API Testing with REST Assured & POSTMAN
- RPA with UiPath
- GitHub Copilot - The Complete Guide - 2026
- Data Structures and Algorithms in Java
- Git & GitHub Bootcamp